



THE CHAI STRAINER AND THE SALT PAN

Learning to Separate Magic Mixtures!

Grade 4 Science Lesson Handbook

 **Kabir and Anya have a muddy playground problem!**

Let's explore how scientists separate messy mixtures using smart tricks!





Kabir and Anya's Rainy Day Challenge

Oh no! A huge rainstorm just hit the neighborhood playground! 🌧️

After a heavy, splashing rain, Kabir and Anya's favorite cricket ground was turned into a total muddy mess! While trying to clean up, their water bucket fell over and got completely mixed up with dirt, brown sand, and salt.

Now they are stuck with a bucket of dirty, salty, muddy brown liquid. They need our help!



The Big Question:

Can you help Kabir and Anya separate this big mess to get clean water, clean sand, and their salt back? Let's learn the secret methods together!

What is a Mixture?

Before we can fix the bucket, we need to know what happened inside it. When you put things together, you create a **mixture**!

A mixture happens when two or more different things are mixed up, but they can still be taken apart or separated back into their original forms.

Let's look at some daily examples of mixtures:

-  **Tea leaves** floating around in hot water.
-  **Brown sand** mixed up in a glass of water.
-  **White sugar** or salt stirred into warm water.

Two Crucial Science Words:

- **Soluble:** Something that can dissolve (melt completely) into liquid and disappear, like salt!
- **Insoluble:** Something that cannot dissolve, so it floats around or sinks, like sand!



How Filtration Works: The Barrier Rule

Our very first trick for taking mixtures apart is called **Filtration**. It works by using a barrier with tiny, microscopic holes!

Imagine a tiny net or a screen. When you pour a mixture through it, the fluid flows right through the holes, but solid blocks that are too big get stopped completely!



The Residue (The Leftovers)

The solid bits that are too big to pass through the filter get trapped and stay behind. This trapped solid is called the **Residue**!



The Filtrate (The Clean Liquid)

The clean liquid that successfully passes through the tiny holes in the filter is called the **Filtrate**!



An Indian Example: The Chai Strainer

You see filtration happen every single day in your kitchen! 🏠

When your parents make hot morning tea, they use a special tool called a "**Chai-ni**" (a tea strainer). This strainer has a net pattern made of tiny gaps.

When the messy tea is poured into the cup:

- 🍃 The strainer stops the big black **tea leaves** from falling into your cup. This trapped pile is the **Residue**!
- ☕ The delicious, smooth **liquid tea** passes right through the net gaps. This clean liquid is our **Filtrate**!

So, a Chai-ni is just a brilliant science filter! 🧠

How Evaporation Works!

But wait! What happens if a solid is **soluble** and dissolves completely, like salt? The salt particles become too tiny to be caught by a filter cloth. They slip right through the holes with the water!

To catch dissolved pieces, we use a different magic trick called **Evaporation**! We use heat instead of a filter cloth.

The Disappearing Act:

When salty water is heated, the liquid water transforms into invisible steam and floats away into the sky. But the salt particles cannot turn into steam! They get left behind at the bottom of the pan as dry white crystals.

 **Safety Note: Heating experiments can be dangerous. Always do heating activities with an adult helper watching you!**

An Indian Example: The Salt Pan

Did you know that the salt you eat on your food is made by the sun?



In sunny coastal places like **Gujarat** and **Tamil Nadu**, salt farmers use natural evaporation on a giant scale! They build huge, flat, shallow ground spaces called **salt pans**.

First, they let salty seawater flow into these wide pans. Then, they let the blazing hot sun heat the water day after day. As the water disappears into steam up into the blue sky, beautiful white salt crystals are left behind on the ground!

The salt farmers simply rake up these leftover crystals to give us our kitchen salt!



When We Need Both Magic Steps!

Now we have all the super science tools to solve Kabir and Anya's muddy water challenge! Because their bucket has both **insoluble sand** and **soluble salt**, we must use both methods in order!

The Action Plan	What Happens?
STEP 1: Filtration	Pour the mixture through a filter cloth. The heavy sand gets TRAPPED on top (Residue). The clear salt water flows right through (Filtrate).
STEP 2: Evaporation	Heat that clear salt water filtrate gently in a flat pan. The water turns to steam and disappears into the air, leaving clean SALT CRYSTALS behind!

Hooray! The bucket is clean, and all elements are separated safely! 🎉



Guided Lab Practice: River Water Cloth Filter

Let's look at how Kabir uses this science at home. Kabir folds a clean piece of **Suti Kapda** (white cotton cloth) and places it tightly over an empty bucket. He slowly pours muddy river water through it...

Let's identify what happens inside the lab:

-  **What gets TRAPPED in the cloth?** Silt (fine mud particles), dirt, small pebbles, and tiny leaf bits. This is our **Residue!**
-  **What PASSES THROUGH the cloth?** Clear, cleaner water is collected down below. This is our **Filtrate!**



The Rann of Kutch Fact:

The famous **Rann of Kutch** in Gujarat is one of the world's largest salt deserts! It works exactly like a giant evaporation lab run by the hot sun!



Summary: Remember Your Rules!

-  ****Insoluble? Filter It!**** If a solid cannot dissolve, filter it to catch it as ****Residue****.
-  ****Soluble? Evaporate It!**** If it dissolves like salt, heat it so water turns to steam and leaves ****Crystals**** behind.



Your Secret Home Mission!

Watch an adult make hot chai tea or filter water at home. Be a science detective!

- Spot the **Residue** trapped behind (like black tea leaves).
- Spot the **Filtrate** passing into the cup (smooth liquid tea).
- Spot the **Evaporation** steam rising from the hot pot!



Quick Check: Can salt be removed by a cloth filter? **No!** It is dissolved and needs heat!

THANK YOU! KEEP EXPLORING! 